05|08|26

Signal Processing

Announcement

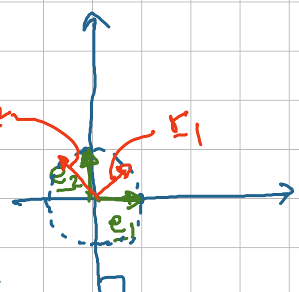
- Quiz tomorrow. 06. 08. 26
- In class, 30 mins
- Closed book, closed notes

- Recap: DFT and notion of frequency
- DFT of $x_1[n]$ and $x_2[n]$
- Discrete-time Frequency Transform (DTFT)
- DTFT of $h_{ave}[n]$ and $h_{diff}[n]$
- Z-transform
- Pause and reflect

- $x_1[n] = \begin{cases} 1 & 0 \le n \le 7 \\ 0 & \text{else} \end{cases}$ $\qquad x_2[n] = \begin{cases} \cos \pi n, & 0 \le n \le 7 \\ 0 & \text{else.} \end{cases}$

- $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k n}{N}}$

- Change of basis in $\mathbb{R}^2$
  - Standard basis in $\mathbb{R}^2$: $\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  - $\underline{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\qquad E = \begin{bmatrix} \underline{e}_1 & \underline{e}_2 \end{bmatrix}$



  - New basis $\underline{r}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ $\quad \underline{r}_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ $\quad R = \begin{bmatrix} \underline{r}_1 & \underline{r}_2 \end{bmatrix}$

  - Project $\underline{x}$ onto the new basis R.

$$x_p = \begin{bmatrix} \underline{r}_1^T \\ \underline{r}_2^T \end{bmatrix} \underline{x} = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix}$$

$$x_p[1] = \langle \underline{r}_1, \underline{x} \rangle$$

$$x_p[2] = \langle \underline{r}_2, \underline{x} \rangle$$

- Now, let's find the DFT of $x_1[n]$, $x_2[n]$ $(N=8)$. $X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j\frac{2\pi k n}{N}}$

- $x_1[n] = [\underset{\substack{\uparrow \\ 0}}{1} \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1] = \underline{x}_1$

Let $\underline{x} = [x[0], \dots x[N-1]]^T$

$\underline{W}_N^k = [e^{j\frac{2\pi k 0}{N}}, \dots e^{j\frac{2\pi k(N-1)}{N}}]^T$

$\Rightarrow X[k] = \langle \underline{x}, \underline{W}_N^k \rangle$

Aside:

$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \cdot e^{j\frac{2\pi n k}{N}}$

$k=0 \quad \underline{W}_N^0 = [1 \ 1 \ 1 \ 1 \ \dots \ 1]$

$k=1 \quad \underline{W}_N^1 = [1 \ e^{j\frac{2\pi}{N}} \ \dots \ e^{j\frac{2\pi(N-1)}{N}}]$

$k=N/2 \quad \underline{W}_N^{N/2} = [1 \ -1 \ 1 \ -1 \ \dots \ -1]$

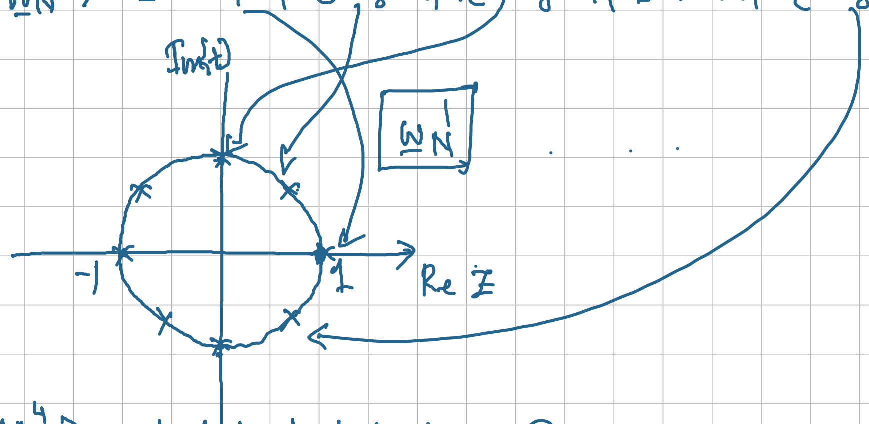
$k=k' \quad \underline{W}_N^{k'} = [1 \ e^{j\frac{2\pi k'}{N}} \ \dots \ e^{j\frac{2\pi k'(N-1)}{N}}]$

$\vdots$

$k=N-1 \quad \underline{W}_N^{N-1} = [1 \ e^{j\frac{2\pi(N-1)}{N}} \ \dots \ e^{j\frac{2\pi(N-1)^2}{N}}]$

$X_1[0] = \langle \underline{x}_1, \underline{W}_N^0 \rangle = 1\cdot1 + 1\cdot1 + 1\cdot1 + 1\cdot1 + 1\cdot1 + 1\cdot1 + 1\cdot1 + 1\cdot1 = 8$

$X_1[1] = \langle \underline{x}_1, \underline{W}_N^1 \rangle = 1 + e^{-j\frac{2\pi}{8}} + e^{-j\frac{2\pi \cdot 2}{8}} + \dots + e^{-j\frac{2\pi 7}{8}} = 0$



$\vdots$

$X_1[4] = \langle \underline{x}_1, \underline{W}_N^4 \rangle = 1-1+1-1+1-1+1-1 = 0$